



AJAYI CROWTHER UNIVERSITY, OYO
FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER ENGINEERING
FIRST SEMESTER EXAMINATION

COURSE TITLE: DIGITAL SYSTEM DESIGN WITH VHDL (ECE 5159)
COURSE CODE: ECE 5159
DURATION: 2¹/₂Hrs

INSTRUCTION: Answer QUESTION ONE and ANY THREE Questions

- 1a. What are the main tasks required in developing a digital system? (2mks)
- 1b. A three-way light control is to be installed in a large room with three doors and a switch near each door controls a light in the room. The switch is turned on or off by changing the state of any one of the switches according to the following conditions:
The light is off if all switches are open; closing any of the switches turns the light on; Turning on a second switch turns the light off; The light will be on if exactly one switch is closed and off if two (or no) switches are closed. If the light is off when two switches are closed, then it should be turned on by closing the third switch. Synthesize the three-way light by
- (i) creating a truth table (5mks)
 - (ii) generating the POS expression and optimization (6mks)
 - (iii) generating the SOP expression and optimization (6mks)
 - (iv) determining the K-map optimization (3mks)
 - (v) the final synthesis (3mks)
- 2a. Describe the two types of state Machines. (4mks)
- 2b. For the Moore machine schematic of a light switch as shown in Figure 1,
- i. determine the corresponding Moore state transition table (3mks)
 - ii. draw the Mealy machine schematic of a light switch (5mks)

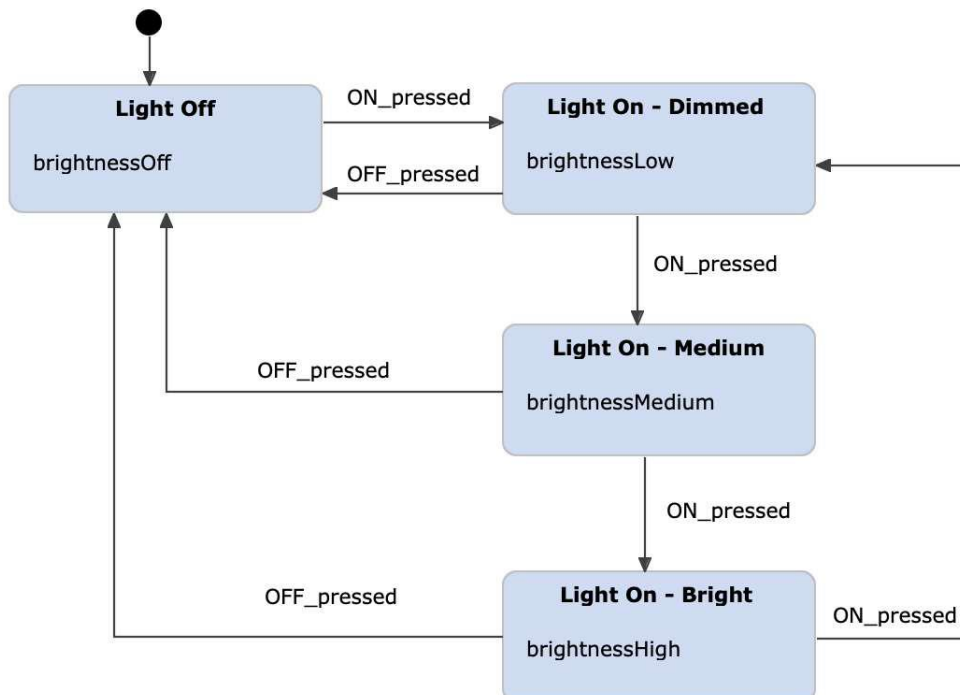


Figure 1

- 2c. Enumerate six features of Harel state chart (3mks)

- 3a. Write a complete VHDL code for the 4-input network circuit in Figure 2, with the ENTITY name as **Network** and the ARCHITECTURE name as **Behaviour** (6mks)

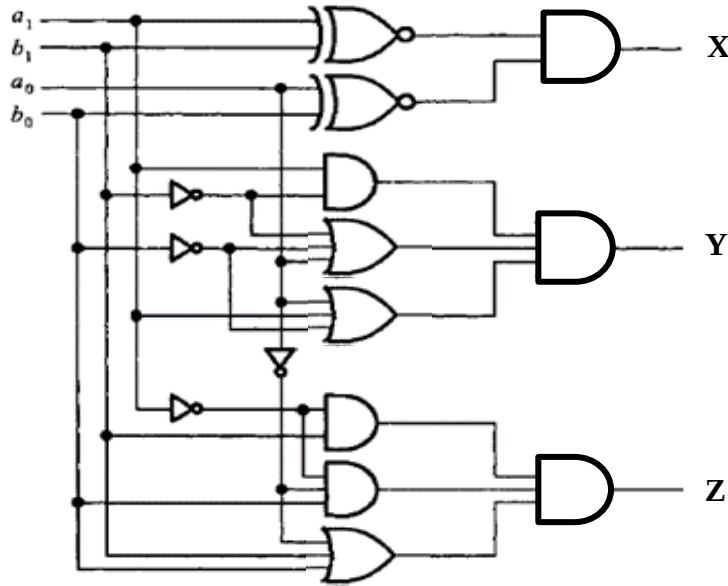


Figure 2

- 3b. Determine the corresponding truth table for the timing diagram on the waveform editor of Figure 3 and express the output in POS term? (4mks)

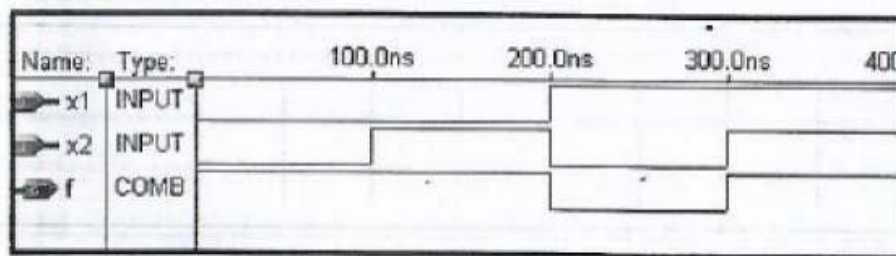


Figure 3

- 3c. Determine f for the Schematic capture of the hierarchical design of Figure 4 using the result from question 3b. (5mks)

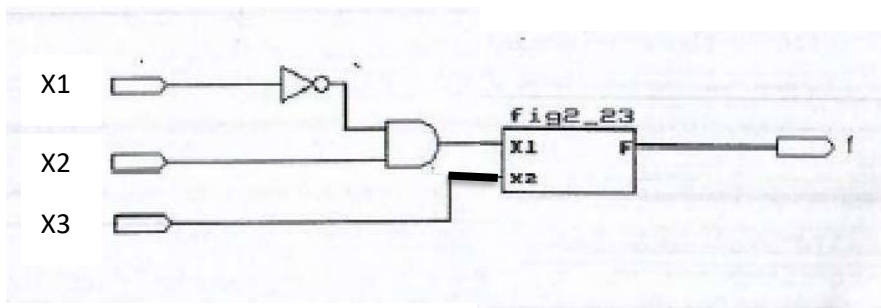


Figure 4

- 4a. Determine the validity of the VHDL data object names in Table 1 and the reasons for assertion in the necessary columns. (4mks)

Table 1

S/N	VHDL data object names	Valid/Invalid	Reason for choice
1	<u>CPEdept</u>		
2	COMPONENT		
3	CPE_5159_		
4	123abc		

5	PACKAGE		
6	CPE5159__exam		
7	ACU		
8	David_5159		

4b. Let A and B be VHDL signals with values A = 0111 and B = 0100, determine the Boolean result **with justification** of the comparison

A < B

for (i) SIGNED type A signal data object

(ii) UNSIGNED type signal data object (5mks)

4c(i). Discuss the concept of resolution function using a schematic diagram (3mks)

(ii). What is the difference between the two signals object declaration below (3mks)

SIGNAL j : BIT_VECTOR (1 TO 4);

SIGNAL Byte : BIT_VECTOR (7 DOWNT0 0);

5a. Determine the error in the following code by rewriting the correct code with the addition or omission Underline (11mks)

LIBRARY ieee ;

ENTITY mux2to1 IS

PORT (d₀, s : IN STD_LOGIC;

Z : OUT STD_LOGIC);

END mux2to1;

ARCHITECTURE Behaviour OF mux2 IS

BEGIN

IF s = '0' THEN

Z <= ;

ELSE

Z <= d₁;

END IF;

END PROCESS;

END Behaviour;

5b. Describe the two assignment statements briefly (4mks)

6a. Design the circuit represented by this RTL statement

A ^ B : R1 <- R2 (7mks)

6b. Write a VHDL code for a 2-to-1 multiplexer circuit using a selected signal assignment statement with inputs - X₀ X₁, output – Z, select pin – n, name of entity – mux2to1 and name of architecture – Function. (5mks)

6c. Differentiate between a Register Transfer Level Design (RTL) and Synchronous Sequential Logic Design (SSL) (3mks)